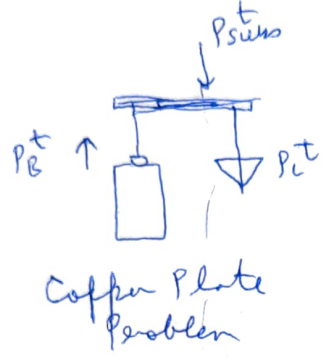


MPOPF Objective Problem



$$\min \sum c^t p_{sub}^t$$

s.t.

NRPB $p_{sub}^t + p_B^t = p_L^t \quad \forall t=1:T$

Soc Lin $B^{t+1} \in [\underline{B}, \bar{B}] \quad \forall t=1:T$

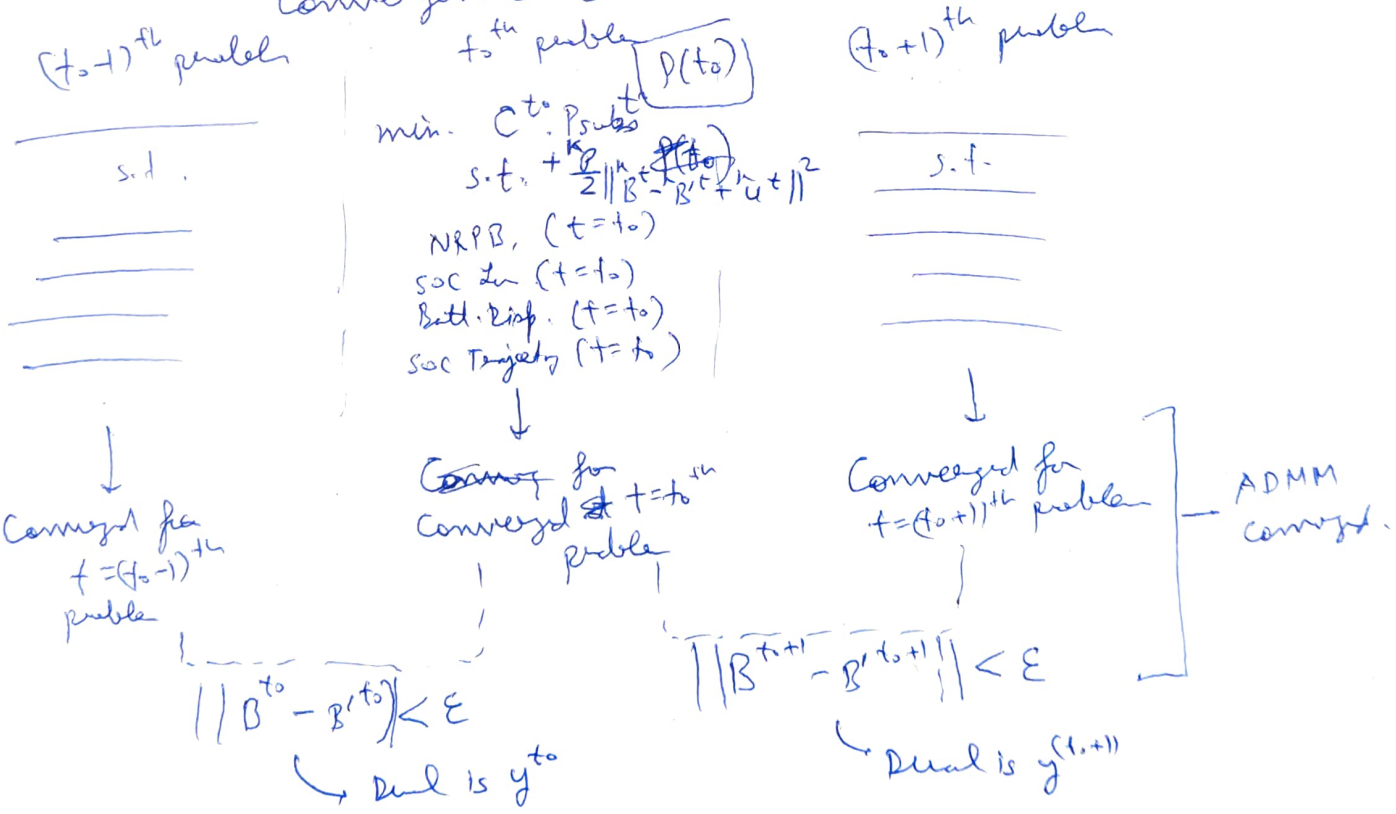
Bath. Disp. $p_B^t \in [-p_{B_R}, p_{B_R}] \quad \forall t=1:T$

Soc Trajectory $B^t = B^{t-1} - \Delta t p_B^t \quad \forall t=1:T$

$B^0 = B_0$ (given)

~~ADMM formulation: ((k+1))th iteration~~

~~I. A Using some~~ ADMM Modelling:
Convergence Criterion:



ADMM Workflow $(k+1)^{th}$ iteration

1.2

~~Have an initial~~

~~get B^0~~

[1]

get B^{k+1} to
like n

using ~~same~~ last iter.

B^{k,t_0}
 u^{k,t_0}
 z^{k,t_0}

B^{k+1,t_0}
 u^{k+1,t_0}

scaled y^{k,t_0}

$\forall t_0 = 1 + T$

solve $p(t)$ $\forall t = 1 + T$
to get B^{k+1,t_0}

[2] get B^{k+1,t_0} using B^{k,t_0} , u^{k,t_0}
 z^{k+1,t_0}

$$B^{k+1,t_0} = \arg \min_{B^{k+1,t_0}} \frac{\mu}{2} \sum \| \cancel{B^{k,t_0}}^{k+1,t_0} B^{k+1,t_0} - B^{k,t_0} + u^{k,t_0} \|^2$$

s.t.

$$B^{k+1,t_0} \in [\underline{B}, \overline{B}]$$

[3] update dual u^{k+1,t_0}

$$u^{k+1,t_0} = u^{k,t_0} + \cancel{B^{k,t_0}}^{k+1,t_0} B^{k+1,t_0} - B^{k,t_0}$$

[4] update P^{k+1} as well.